

Epic 4: Model Building

Story 3: K-Nearest Neighbours (KNN) Model

The K-Nearest Neighbours (KNN) algorithm is implemented to classify flood risk based on the similarity between weather-related features. A reusable function named `KNN()` is created to train, test, and evaluate the model using the training and testing datasets.

KNN Algorithm

The `KNeighborsClassifier` from the Scikit-learn library is used to build the classification model. KNN is a supervised machine learning algorithm that classifies a new data instance by identifying the K nearest neighbors in the training dataset and assigning the majority class among them.

Model Initialization

The classifier is initialized with the parameter:

- `n_neighbors = 5` – Specifies that the prediction is based on the five nearest neighboring data points.

Choosing an appropriate value for K helps balance prediction accuracy and model stability.

Model Training

The model is trained using the `fit()` method with the training dataset (`X_train` and `y_train`). During training, the algorithm stores the training data and learns the relationship between the input features and the target variable.

Prediction

After training, the `predict()` method is applied to the testing dataset (`X_test`) to generate flood predictions. The predicted values are then compared with the actual target values (`y_test`) to evaluate the model's performance.

Model Evaluation

The performance of the KNN model is evaluated using the following metrics:

- **Accuracy Score** – Measures the overall percentage of correctly classified instances.
- **Confusion Matrix** – Displays the number of True Positives (TP), True Negatives (TN), False Positives (FP), and False Negatives (FN).
- **Classification Report** – Provides detailed evaluation metrics including Precision, Recall, F1-Score, and Support for each class.

Outcome

- Successfully trained the K-Nearest Neighbours classifier.

- Generated flood predictions using the testing dataset.
- Evaluated the model using Accuracy, Confusion Matrix, Precision, Recall, and F1-Score.
- Compared the KNN model with Decision Tree, Random Forest, and XGBoost to identify the most accurate flood prediction model.

Figure 14: K-Nearest Neighbours (KNN) model training and performance evaluation using the flood prediction dataset.

```

1  from sklearn.neighbors import KNeighborsClassifier
2  from sklearn.metrics import confusion_matrix, classification_report, accuracy_score
3
4  def KNN(X_train, X_test, y_train, y_test):
5      print("\n===== KNN MODEL BUILDING =====")
6
7      # Initialize KNN classifier
8      model = KNeighborsClassifier(n_neighbors=5)
9
10     # Train the model
11     model.fit(X_train, y_train)
12     print("[INFO] KNN model training completed.")
13
14     # Predict on test data
15     y_pred = model.predict(X_test)
16     print("[INFO] Prediction completed on test data.")
17
18     # Evaluate model performance
19     accuracy = accuracy_score(y_test, y_pred)
20     cm = confusion_matrix(y_test, y_pred)
21     cr = classification_report(y_test, y_pred)
22
23     # Display results
24     print("\n[RESULT] Accuracy:", accuracy)
25     print("\nConfusion Matrix:")
26     print(cm)
27     print("\nClassification Report:")
28     print(cr)
29
30     # Return model and predictions
31     return model, y_pred
32
33 # Example Usage
34 # knn_model, knn_pred = KNN(X_train, X_test, y_train, y_test)

```